

HPPN: High Pulsed Plasma Nitriding

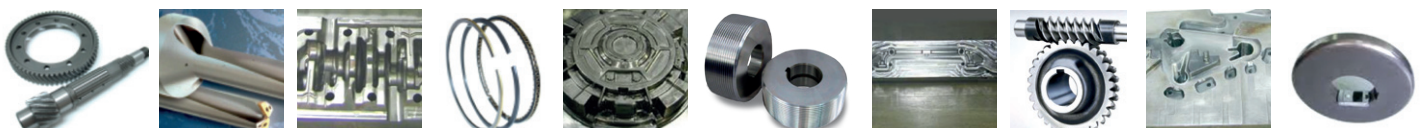
EASY AND EFFICIENT:
THE WAY PLASMA
NITRIDING SHOULD BE

HPPN PLASMA NITRIDING ADVANTAGES:

- Extremely low running cost
- Fast heating and homogeneous temperature
- Flexible and robust technology
- Control of metallurgical structures



THE MOST COST EFFICIENT TECHNOLOGY



DUPLEX DLC TECHNOLOGY : Plasma Nitriding + DLC

New Technology

High Pulsed Plasma Nitriding Benefits

The HPPN power supply offering a wide frequency range from 5 to 150 kHz:

- ensures no substrate damage due to sputtering effect (ex: high polish surface)
- minimizes substrate heating, keeps an homogeneous temperature in the batch
- nitriding in post discharge
- controlled structure
- High reliability.

Quick Heating by radiation ; and fast cooling to reduce cycle time.

A plasma screen is obtained by using specifically designed shields, placed around the parts which quickly heats up the parts in a neutral gas.

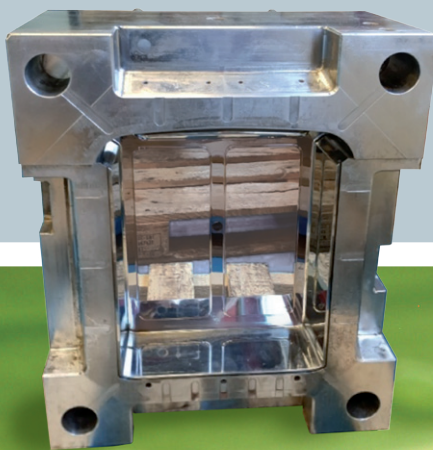
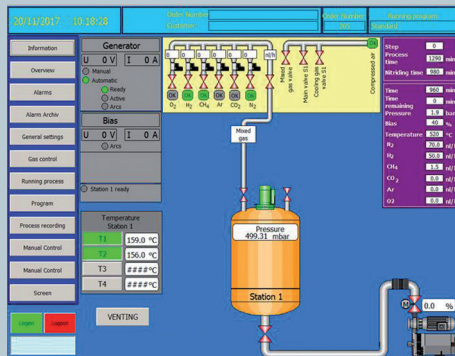
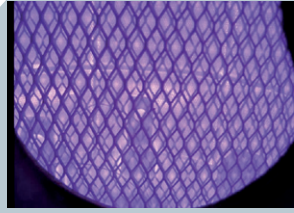
At the end of the cycle, fast cooling is applied by forced convection in an inert gas.

Automatic control

The electronic control system provides fully automated supervision of the process. For example, it controls and monitors:

- heating ramp up and regulation
- the temperature of the work load
- the partial pressure of process gas
- the gas mixture
- the different process steps and process time

In addition, it also checks all safety features and alarm defects. The fixed thermocouple dummy in the anode prevents the parts from overheating.



Brilliant Nitriding Process

Special Processes advantages

Nitriding of Stainless Steel:

S phase: 1400 HV keeping the corrosion resistance

Brilliant Nitriding Process:

To protect high polish molds: keep the roughness and the brilliance shiny surface.

Typical Applications

- components: crankshafts, camshafts, gears, pistons, valves, cylinders, valve springs, shafts, spindle, sliding rails, pump cylinders
- all types of hot forging dies
- all types of extrusion dies
- sheet metal forming punches and dies
- rolling dies
- aluminium diecasting moulds
- plastic extruder screws, moulds for injection moulding
- tool holders and taps.

Machines are available in different sizes and can accommodate high volume production. We manufacture customized solutions.

PD2i has service and spare parts centers in Europe, Asia and Americas.

"...you won't believe all we have on offer"

PLASMA SURFACE TECHNOLOGIES

PLASMA NITRIDING, DUPLEX DLC, DLC



	High Pulsed Plasma Nitriding	Duplex DLC	MpC DLC 700
Available Technologies	Active Screen	Hybrid Plasmanitriding + PeCVD	Hybrid PVD-PeCVD, Duplex DLC
Usefull volume	Ø 600 mm up to Ø 1500 mm H 700 mm up to 2000 mm	Ø 650 mm x 700 mm	Ø 650 mm x 650 mm
Cycle times	Depending on nitriding depth	Depending on nitriding depth	4 - 8 h
Process	Plasmanitriding	Plasmanitriding, Duplex DLC	a-C:H:Me DLC, a-C:Me DLC (hydrogen free) Duplex DLC
Properties	High heat resistance, fatigue and wear resistance	Fatigue and wear resistance, low friction, corrosion resistance	Low friction, corrosion resistance, fatigue and wear resistance
Applications	Automotive, aircraft, hydraulic, textile, molds & dies, tooling	Automotive, aircraft, hydraulic, textile, plastics molds, tooling	Automotive, aircraft, hydraulic, textile, plastics molds, tooling

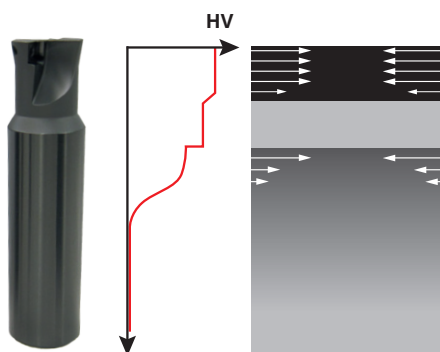


Duplex DLC: A unique process in a single piece of equipment

It is a PeCVD process in Plasma nitriding equipment, our HPPN power supply offers the possibility to make PECVD after the nitriding process without any process interruption.

Advantages:

- 1 piece of equipment, 2 surface treatments
- better fatigue and wear resistance
- low friction and improve lubricity
- good corrosion resistances
- short cycle, very low running costs



Example:

Replacement of nitriding + black oxide by DUPLEX DLC on Tool Holders

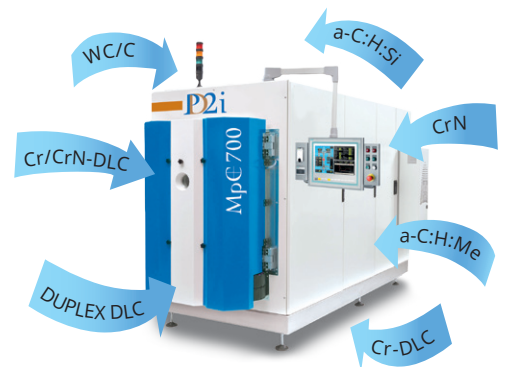
DLC 5µm • nitriding layer 872 HV • 0,15 mm

Benefits:

- reduced lead time (1 week)
- reduced production costs (~30 %)
- better coefficient of friction

DLC Equipment: 100 % tailored to customer needs

Select the technology which offers the best performance for your application.



A GLOBAL NETWORK TO SUPPORT OUR CUSTOMERS



YOUR PARTNER WHEN IT IS ALL ABOUT PERFORMANCE

Plasma ion nitriding offers one of the most advanced and environmentally-friendly heat treating processes. The economic and environmental benefits combined with our process know-how, developed over decades, will support your business.



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